

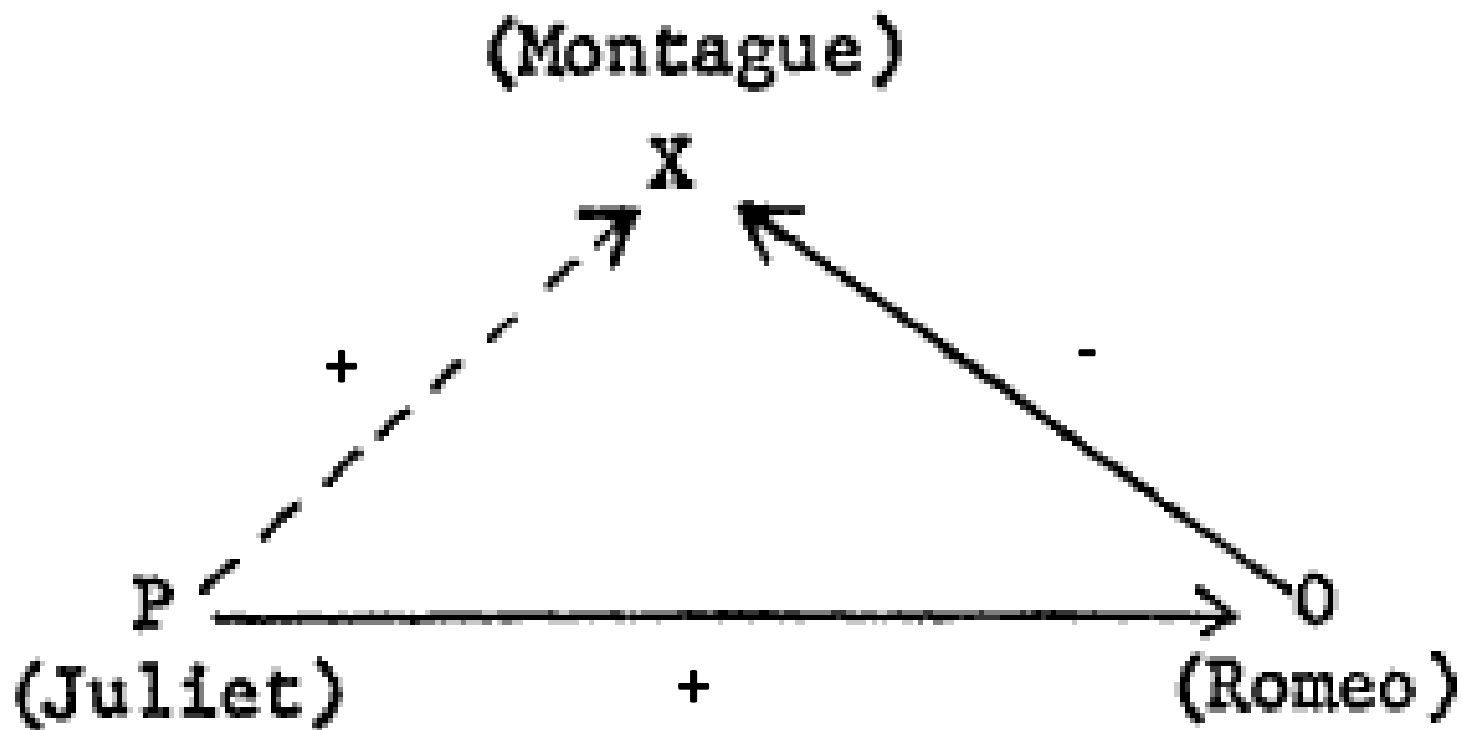


FIG. 1.—The *P-O-X* cycle

(Republican Party)

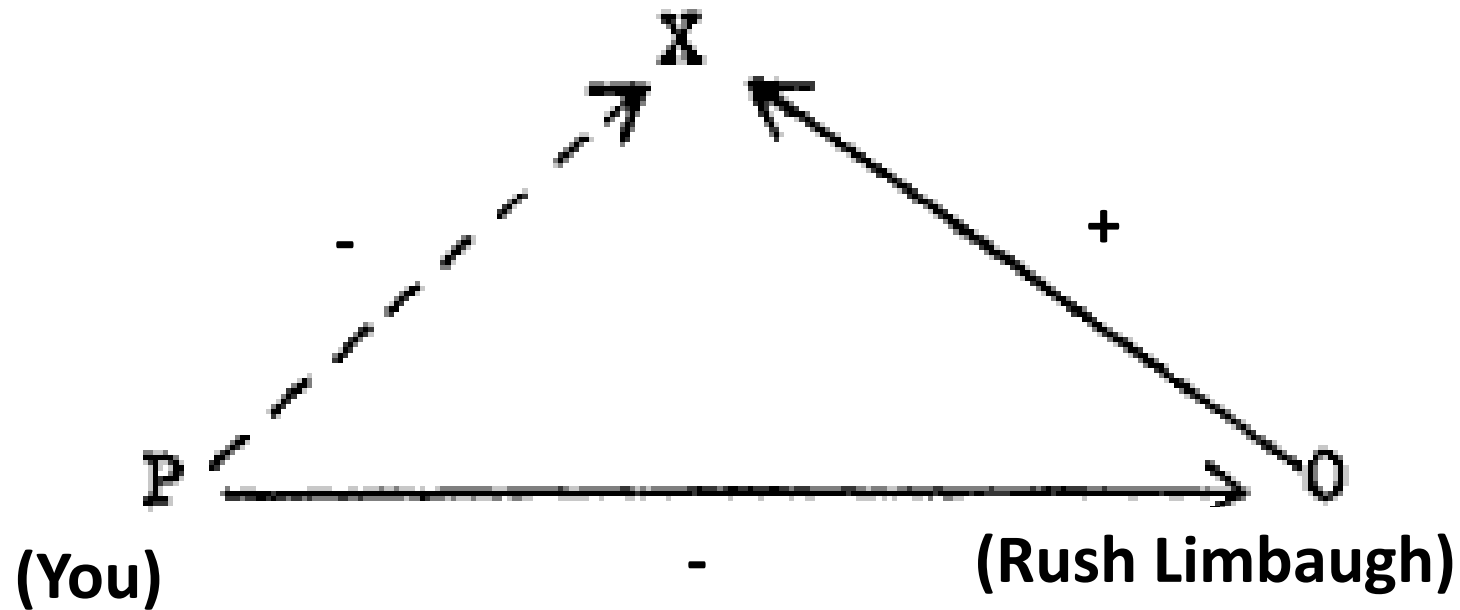


FIG. 1.—The *P-O-X* cycle

PO		OX		PX
+	*	+	=	+
-	*	+	=	-
+	*	-	=	-
-	*	-	=	+

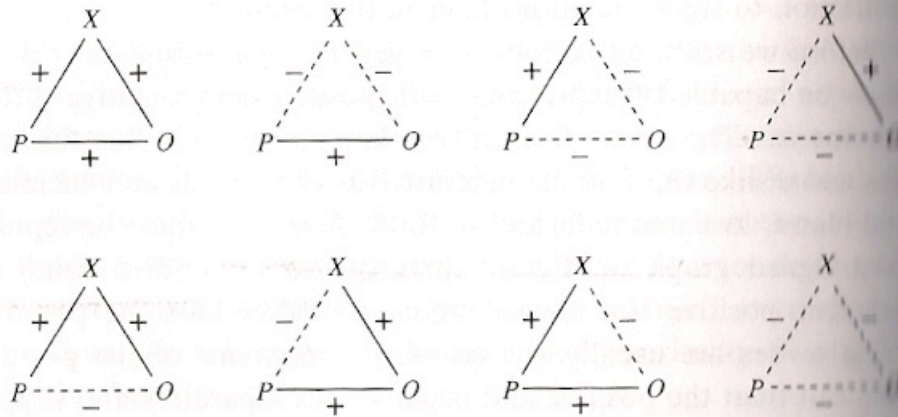


Fig. 6.1. The eight possible $P-O-X$ triples

1. Love anyone your friend loves
2. Hate anyone your friend hates
3. Hate anyone your enemy loves
4. Love anyone your enemy loves

Key result

- Cartwright and Harary (1956) proved that:
 - If all of the triads in a signed value matrix are balanced, then:
 - The matrix can be blocked into two mutually exclusive groups such that:
 - 1) All edges with positive weights are *within* blocks
 - 2) All edges with negative weights are *between* blocks.
 - Formalizes a key intuition between balance processes at the micro-level and macro-level group structure.

PERSON, OTHER, AND X

D1. The more similar *Person* is to *Other*, the more *Person* will like *Other*.

Def. 15. *Person* and *Other* are peers if r_{PO} is positive.

D2. The value of *P*'s liking for *O* tends to be the same as the value of *O*'s liking for *P* (i.e., liking tends to be symmetrical).

Def. 16. If *P* likes *O* and *O* likes *P*, *P* and *O* are friends.

D3. Friends tend to become similar in activities.

D4. Friends tend to become similar in attitudes.

D5. If P and O are friends but differ in their degree of liking, the one with the greater degree of liking will imitate the one with the lesser, rather than vice versa.

Although liking tends toward symmetry, it need not be completely so. If there is a discrepancy, the individual with the greater liking has the greater (PO) value, and the greater pressure to seek similarity vis-à-vis X , an idea somewhat like Willard Waller's "Principle of Least Interest."

D6. If the relationships between *Person* and *Other* and between *Other* and *X* (where *X* is a third individual) are both marked by dislike, the relationship between *Person* and *X* may be friendly.

D7. If *Person* likes *Other* and *Other* dislikes *X*, *Person* will tend to dislike *X*.

D8. If *Person* likes *Other* and *Other* likes *X*, *Person* will tend to like *X*.

D9. If *Person* dislikes *Other* and *Other* likes *X*, *Person* will tend to dislike *X*.

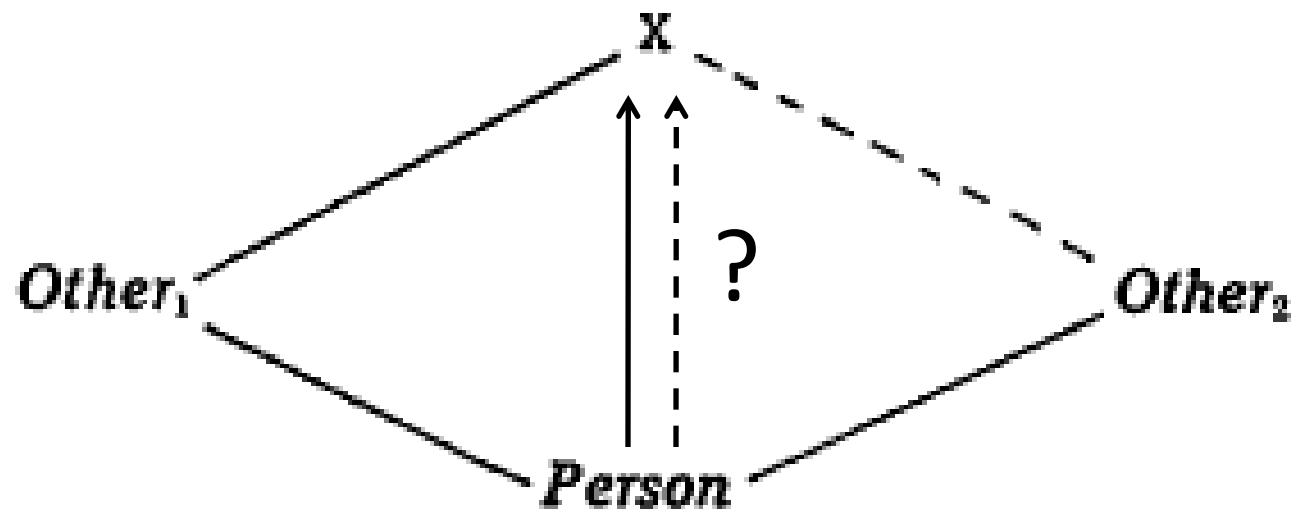


FIG. 2.—The cross-pressure situation

- D11. The greater the cross-pressuring for *Person* (i.e., the more equal the opposite values), the weaker his attitude (liking) toward *X*.
- D12. If *Person* does adopt a positive or negative attitude toward *X* it will be the attitude of the *Other* with the strongest degree of liking or disliking toward *X*, provided that *Person's* bonds to the two groups are equal.
- D13. If *Person* does adopt a positive or negative attitude toward *X*, it will be that of the *Other* to whom he has the stronger positive tie (liking or similarity) providing that the strengths of the *Others'* attitudes toward *X* are equal.
- D14. If *Person* does adopt an attitude toward *X*, he will tend to lower his liking for the *Other* with whom he is now in imbalance and increase his liking of the *Other* with whom he is now in balance.

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- Since triads concern relations among social actors, most often they are studied using a social network formalization. Indeed, triads have been at the heart of theoretical and methodological advances in social network research for nearly half a century. Triads are implicated in many social network theories...

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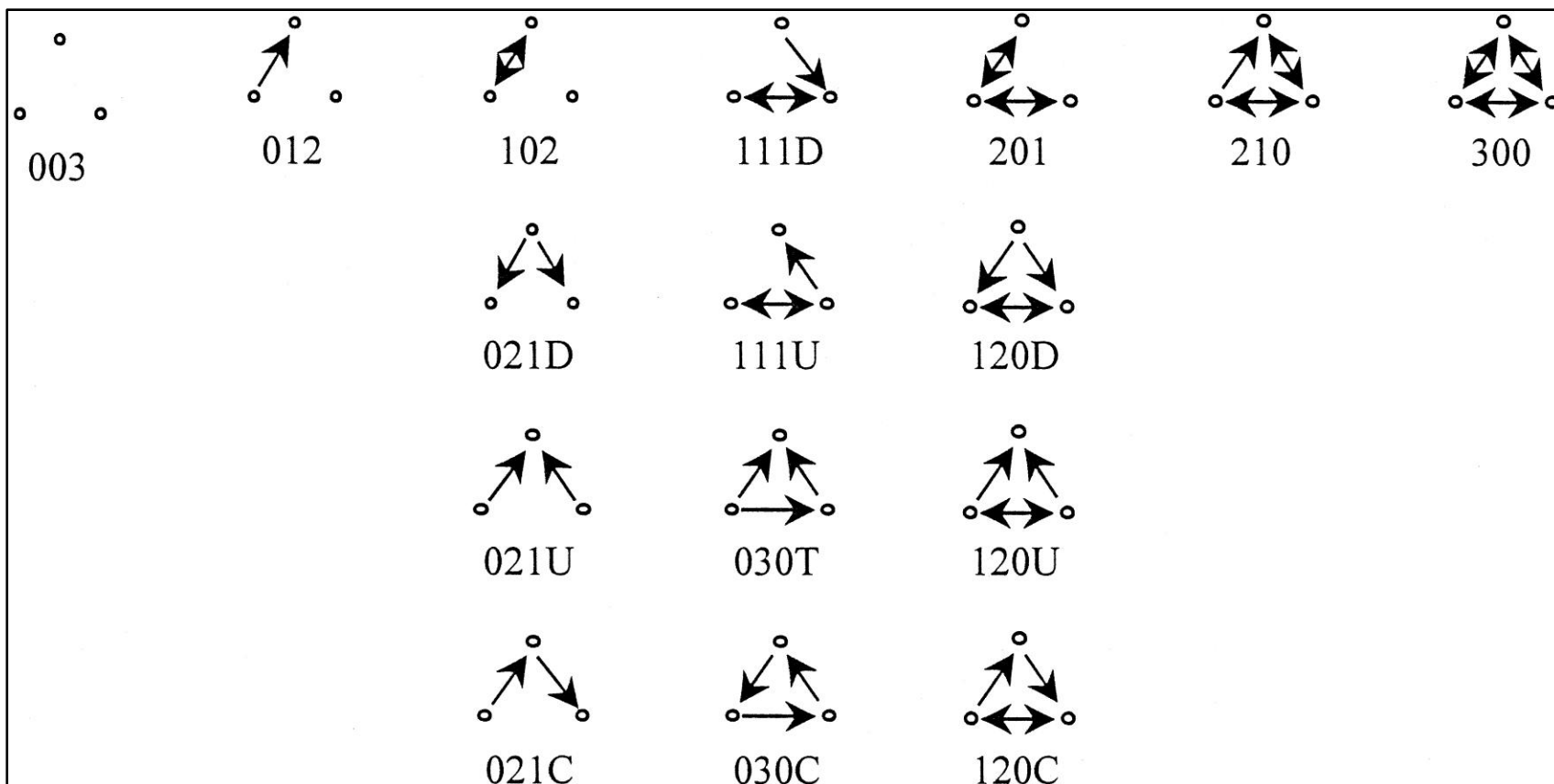
- As introduced in papers by [Davis \(1970, 1977, 1979\)](#) and [Holland and Leinhardt \(1970, 1971, 1972, 1973, 1976, 1979\)](#), and elaborated in the work of [Johnsen \(1985, 1986, 1989a, 1989b, 1998\)](#), **specific global structural patterns imply the presence and absence of specific triadic configurations, just as the occurrence and nonoccurrence of specific triadic patterns imply specific global structural configurations.**

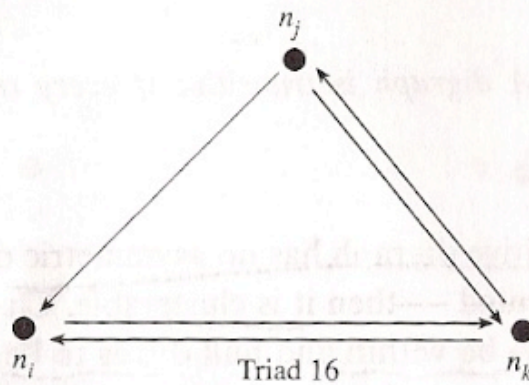
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- Thus, the triad census is useful for evaluating theories about linkage between local processes and global structures because some theoretical global structures are contradicted by specific configurations of **triads**. Such a global theory is evaluated by examining empirical networks for substantial occurrence of triads inconsistent with the theory.

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- *How much* of the triadic structure is accounted for by nodal and dyadic properties? This paper provides an answer to that question and shows that more than 90% of the triadic structure in a collection of diverse social networks is accounted for by lower-order properties. The implications of this result are far-reaching. If a substantial portion of triadic structure is explained by nodal and dyadic features, then the theoretical and methodological importance of *triadic* properties and processes are brought into question.





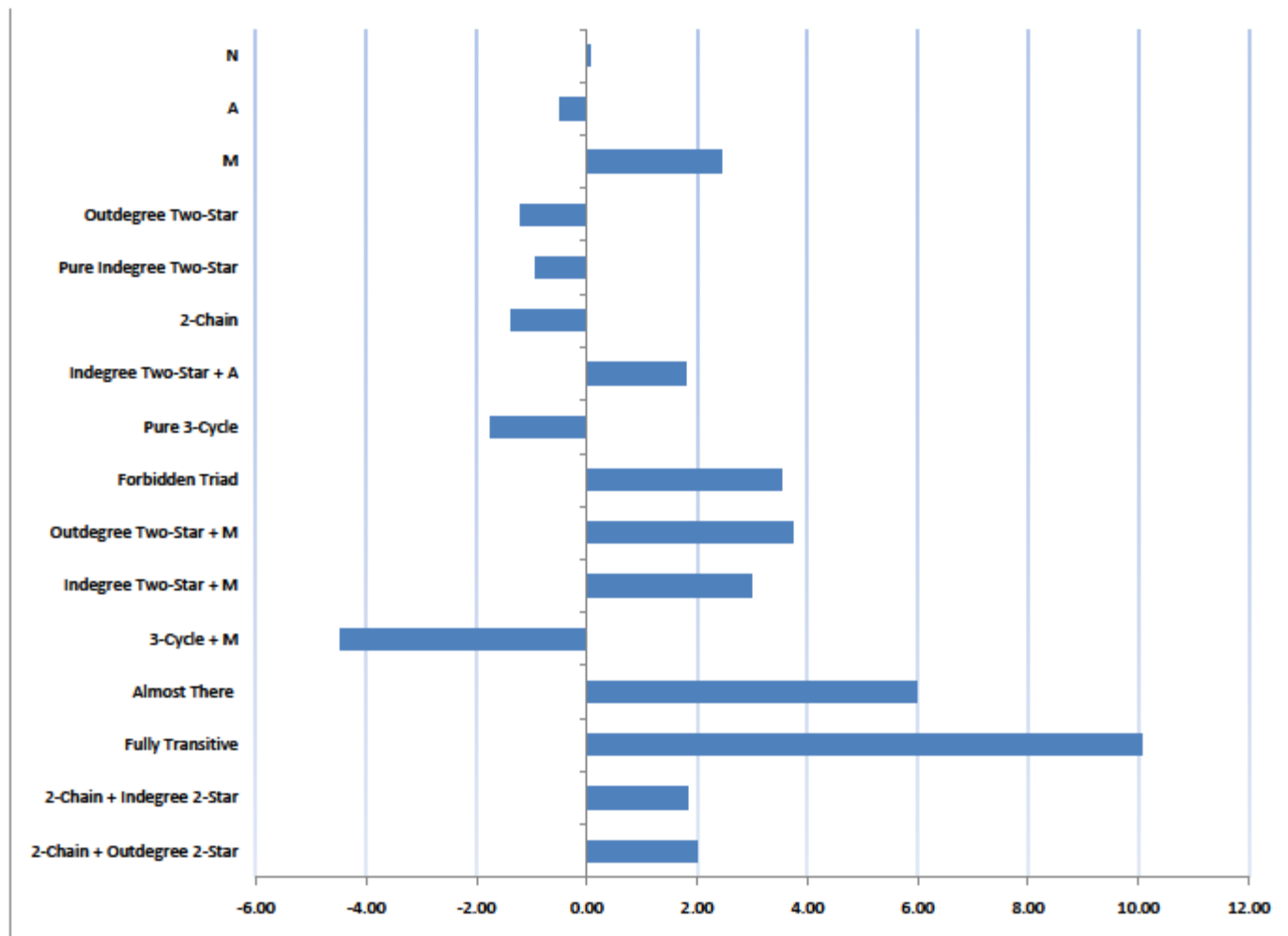
Triple #1 : $n_i n_j n_k$	$n_i \not\rightarrow n_j$	$n_j \rightarrow n_k$	$n_i \rightarrow n_k$	Vacuously transitive
Triple #2 : $n_i n_k n_j$	$n_i \rightarrow n_k$	$n_k \rightarrow n_j$	$n_i \not\rightarrow n_j$	Intransitive
Triple #3 : $n_j n_i n_k$	$n_j \rightarrow n_i$	$n_i \rightarrow n_k$	$n_j \rightarrow n_k$	Transitive
Triple #4 : $n_j n_k n_i$	$n_j \rightarrow n_k$	$n_k \rightarrow n_i$	$n_j \rightarrow n_i$	Transitive
Triple #5 : $n_k n_i n_j$	$n_k \rightarrow n_i$	$n_i \not\rightarrow n_j$	$n_k \rightarrow n_j$	Vacuously transitive
Triple #6 : $n_k n_j n_i$	$n_k \rightarrow n_j$	$n_j \rightarrow n_i$	$n_k \rightarrow n_i$	Transitive

Fig. 6.8. The type 16 triad, and all six triples of actors

Meaningful Label	obs(p)	E(p)	obs/E	ln(obs/E)
2-Chain + Outdegree 2-Star	0.00258	0.00035	7.31	1.99
2-Chain + Indegree 2-Star	0.00219	0.00035	6.19	1.82
Fully Transitive	0.00010	0.00000	23428.58	10.06
Almost There	0.00025	0.00000	396.01	5.98
3-Cycle + M	0.00010	0.00908	<i>0.01</i>	-4.47
Indegree Two-Star + M	0.00015	0.00001	19.52	2.97
Outdegree Two-Star + M	0.00031	0.00001	41.83	3.73
Forbidden Triad	0.00025	0.00001	33.47	3.51
Pure 3-Cycle	0.00002	0.00012	<i>0.18</i>	-1.73
Indegree Two-Star + A	0.00214	0.00035	6.07	1.80
2-Chain	0.00214	0.00836	<i>0.26</i>	-1.36
Pure Indegree Two-Star	0.00166	0.00418	<i>0.40</i>	-0.92
Outdegree Two-Star	0.00127	0.00418	<i>0.30</i>	-1.19
M	0.04862	0.00418	11.63	2.45
A	0.12196	0.19779	<i>0.62</i>	-0.48
N	0.81625	0.78010	1.05	0.05

Table 1
Triad types

Triad type		Probability in Bernoulli digraph
003		$(1 - d)^6$
012		$6d^1(1 - d)^5$
102		$3d^2(1 - d)^4$
021D		$3d^2(1 - d)^4$
021C		$6d^2(1 - d)^4$
021U		$3d^2(1 - d)^4$
111U		$6d^3(1 - d)^3$
030T		$6d^3(1 - d)^3$
030C		$2d^3(1 - d)^3$
111D		$6d^3(1 - d)^3$
201		$3d^4(1 - d)^2$
120U		$3d^4(1 - d)^2$
120C		$6d^4(1 - d)^2$
120D		$3d^4(1 - d)^2$



$$\Delta=\frac{\sum\limits_{i=1}^gx_{i+}}{g(g-1)}=\frac{\sum\limits_{j=1}^gx_{+j}}{g(g-1)}=\frac{\sum\limits_{i=1}^g\sum\limits_{\substack{j=1\\j\neq i}}^gx_{ij}}{g(g-1)}.$$

$$\Delta=\frac{d}{g-1}.$$

$$\binom{g}{2}=\frac{g(g-1)}{2}$$

$$\binom{g}{3}=g(g-1)(g-2)/6$$

$$t=(c_{003},c_{012},c_{102},c_{021D},c_{021U},c_{021C},c_{111D},c_{111U},c_{030T},c_{030C},\\c_{201},c_{120D},c_{120U},c_{120C},c_{210},c_{300})$$

$$\frac{\sum_{k=1}^K \sum_{j=1}^K \sum_{i=1}^K x_{ij} x_{jk} x_{ik}}{\sum_{k=1}^K \sum_{j=1}^K \sum_{i=1}^K x_{ij} x_{jk}}.$$

From Bonacich and Lloyd (2004)

Balance theory		Status
1	Friends of friends are friends	A positive connection with a high status individual increases ones status
2	Friends of enemies are enemies	A positive connection to a disvalued individual decreases ones status
3	Enemies of friends are enemies	A negative relation to a high status individual reduces ones status
4	Enemies of enemies are friends	A negative relation to a disvalued individual increases one status

Another reason why SVD is your friend

- **Proposition 1**

The eigenvector of a balanced symmetric matrix of affective relations reveals the clique structure.

Name	Status
Clique	A
Romul	0.174
Bonaventu	0.188
Ambrose	0.248
Berthold	0.319
Peter	0.42
Louis	0.219
Victor	0.365
Clique	B
Winfried	-0.081
John	-0.142
Gregory	-0.292
Hugh	-0.088
Boniface	-0.123
Mark	-0.217
Albert	-0.072
Amand	-0.03
Basil	-0.254
Elias	-0.282
Simplicius	-0.287